LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	AC systems - schematic of external power supply, auxiliary power units
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	AC systems - schematic of external power supply, auxiliary power units. Practical on 6-Pin Receptacle system. Introduction to Power Quality analysis (PQA)
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the A/C power systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on 6-Pin Receptacle system. Safety precautions: Verify power off, wear PPE. Instructor supervision: Monitor connections and voltage checks.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary sources of AC power for an aircraft?
(b) Define the function of an Auxiliary Power Unit (APU).
(c) Identify the pin configuration of the standard 6-Pin Receptacle system.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Schematic of External Power Supply • Connection sequence for Ground Power • Interlock mechanisms in power systems • Voltage and frequency standards • Protection devices in AC systems	L & Ds	PPT, IB	15 min
(b)	• Auxiliary Power Unit (APU) Schematic • APU start sequence and power transfer • APU generator control logic • APU shutdown procedures • APU maintenance access points	L & Ds	PPT, PDF	20 min
(c)	• 6-Pin Receptacle Practical Demonstration • Power Quality Analysis (PQA) Introduction • Harmonic distortion basics • Voltage fluctuation monitoring • Frequency stability checks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the interlock mechanism in preventing simultaneous power sources.
(b) Describe two key parameters monitored during Power Quality Analysis (PQA).
(c) List the safety steps required before connecting an external power supply.

12. Home Assignment and Follow Up	Prepare a brief note on the need for static converting equipment (rectifiers, inverters) as covered in the next lesson plan (LP 08), focusing on why AC systems require conversion in specific aircraft zones.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need and types of Static converting equipment - rectifiers, transformers, transformer rectifier units, rotary inverters, static inverters and automatic change over system. Practical on inverter, transformer & rectifier.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min